

CANONICAL LESSON

Quality Is a Decision Variable

Decision-centred foundations for accountable sector intelligence

Production Unit	PU-B07-C03
Notebook	V7_0_N03
Status	Controlled educational draft v0.1
Data boundary	Synthetic instructional data only

Central decision: Judge data fitness for the intended decision across completeness, timeliness, coverage, coherence, revisions, and drift.

Learning outcomes

- Measure fitness for use across relevance, accuracy, timeliness, accessibility, coherence, coverage, representativeness, revisions, and drift; route exceptions to accountable review.

Why this matters

A composite quality average can conceal a failed critical field, unlawful source, unusable delay, or severe subgroup gap.

The analytical workflow is useful only when evidence, uncertainty, authority, and action are connected explicitly. The model supports professional judgement; it does not replace accountable institutional authority.

1. Completeness is decision-specific

- A missing value rate is not enough.
- Identify which fields are mandatory for which decision and whether missingness is concentrated in a subgroup, geography, or period.

2. Timeliness versus decision horizon

- Data can be accurate but arrive too late.
- The timeliness gate compares reporting delay with the period in which intervention remains possible.

3. Weighted coverage

- Record counts can hide undercoverage when missing units represent larger populations.
- Compare unweighted and design/size-weighted coverage.

4. Revisions and reproducibility

- Official values may be revised.
- Preserve vintages, reasons, and dissemination status rather than silently overwriting earlier values.

5. Drift is a review signal

- A distribution change may represent real change, measurement change, coding change, or error.
- It triggers investigation; it does not diagnose the cause.

6. Quality gate and exception workflow

- A composite score helps routing but must not conceal failing critical dimensions.
- Critical failures override the average.

Worked cross-sector example

Weighted coverage reveals when a small number of missing reporting units represent a large share of the affected population.

Boundary: The example supports learning and review. It does not constitute an official statistic, eligibility determination, enforcement action, clinical judgement, or operational authorization.

Practice and reflection

- 1.
- Add a subgroup completeness check.
- 2.
- Explain why averaging can be unsafe.
- 3.
- Propose a revision policy.
- 4.
- Distinguish concept drift from data drift.

Answer key

- 1.
- Group by the policy-relevant subgroup and compare both unweighted and weighted missingness with disclosure controls.
- 2.
- A high score can mask a failed key, unlawful source, or unusable delay; critical gates must override averages.
- 3.
- Preserve vintages, reasons, approval, release status, notification, and reproducible recalculation.
- 4.
- Data drift changes input distributions; concept drift changes the relationship between inputs and the target/outcome.

Operational readiness checklist

- Purpose and decision owner are named.
- Evidence definitions, provenance, and lawful access are documented.
- Uncertainty and abstention are visible.
- Human review, escalation, monitoring, incident response, and retirement are assigned.
- Equity, accessibility, security, and sector-specific approval are complete.